

GORGO: Online Tuning for Cross-Region Network-Aware LLM Serving

Anonymous Author(s)

May 5, 2026

Abstract

In cross-region LLM serving, time-to-first-token (TTFT) is highly sensitive to per-request routing decisions. KV-cache reuse across replicas can eliminate much of the prefill computation on multi-turn workloads, but realizing this potential requires jointly balancing cache locality against replica load and network cost. Additionally, public datasets to evaluate these policies do not represent long-context, high prefix-reuse workloads: LMSYS and WildChat average 2 and 2.5 turns per conversation, 0.5 and 2.9k tokens/request, and 5 and 9% intra-user reuse, respectively. To capture the setting where a majority of requests contain reusable prefixes, we evaluate common routing policies such as least-load, prefix-aware, session-affinity, and others on traces from a production inference endpoint featuring multi-turn conversations (~ 82 requests/user), $7\text{--}45\times$ longer prompts (21k avg tokens), and $2\text{--}6\times$ higher KV-cache reuse (55%). We evaluate policies on a multi-region GPU cluster where cross-region network latency ranges from 26ms to 466ms across replicas to test GORGO, a policy that jointly optimizes cache reuse, network latency, and queue depth with weights optimized online to minimize p95 TTFT with no offline profiling. GORGO achieves the lowest TTFT at every percentile (p50, p95, p99) across both a tuning window and a held-out evaluation window against all other load balancing policies, highlighting the impact of online joint optimization.

1 Introduction

Modern LLM chat services serve interactive workloads where perceived latency is dominated by time-to-first-token (TTFT). On a single replica, TTFT is the sum of three terms: (i) the network round-trip from the client proxy to the replica, (ii) queueing delay behind in-flight requests, and (iii) prefill computation for the input prompt. Inference engines such as SGLang [Zheng et al., 2024b] and vLLM [Kwon et al., 2023] expose radix-tree prefix caches that let an incoming request skip prefill for any token prefix already cached on the handling replica. When prompts are long and prefixes are widely reused, this saving dom-

inates: a 20,000-token prompt that hits a 90% prefix match has only $\sim 2,000$ tokens to prefill, reducing TTFT by an order of magnitude.

The KV-cache reuse opportunity is contingent on routing. If a request is sent to a replica that has not seen its prefix, the saving is forfeit. If many requests pile onto the same warm replica, that replica becomes a queueing bottleneck and the cache hit no longer pays for itself. If the closest replica with a warm cache is on the other side of the world, the network round-trip can swallow what the cache hit was supposed to save. The routing layer therefore has to balance three terms simultaneously: network distance to the replica, current load, and reusable cache state for the incoming prompt.

Standard heuristics—least-request, least-load, prefix-aware routing, session-affinity—each optimize one of these signals and ignore the others. For workloads with weak prefix reuse and uniform replica latencies, these heuristics work well [F5 NGINX, 2024]. Interactive long-context traffic breaks both assumptions: production chat traces concentrate most tokens in a small set of returning users with substantial prefix overlap, and replicas placed in different regions have round-trip times that differ by an order of magnitude. The right routing target for a given request can change minute-to-minute as load shifts, and weighting the three terms with static rules leaves performance on the table.

Two further factors complicate empirical evaluation. First, the public chat datasets used in prior LLM-serving evaluations underrepresent long-context multi-turn traffic with reusable prefixes. LMSYS-Chat-1M [Zheng et al., 2024a] and WildChat-4.8M [Zhao et al., 2024] average 470 and 2,900 tokens per request, with intra-user prefix reuse below 10%. A production endpoint we evaluate carries 21,000 tokens per request on average and 53.7% intra-user prefix reuse—a regime where joint cache-network-queue trade-offs dominate TTFT. Second, evaluation methodology matters: a routing policy that wins p50 by herding traffic onto a single warm replica typically loses tail latency once that replica saturates. Reporting only one percentile hides this trade-off.

Contributions.

- A workload characterization (§2) placing three datasets on the same prefix-reuse and request-length axes. The production GLM-5.1 trace has $7\text{--}45\times$ longer prompts and $2\text{--}6\times$ higher token-weighted reuse than LMSYS-Chat-1M and WildChat-4.8M.
- A routing cost model and three variants collectively called GORGO (§3), which score each replica by an additive combination of EWMA network RTT, estimated prefill cost over uncached tokens, and a queue-depth penalty. Variants differ only in how the two scalar weights are set: a static configuration, a per-replica online fit, and a Gaussian (1+1)-evolution-strategy hill climb that directly minimizes p95 TTFT.
- An experimental setup (§5) running each of eight policies on its own dedicated three-replica fleet of L40S:2 SGLang servers in Asia, Europe, and the US East coast, with cross-region RTTs spanning 26–466 ms.

Table 1: Workload characterization. Token counts use the same byte-pair tokenizer in all rows. Reuse percentages are token-weighted prefix-trie savings as described in §2.

Dataset	Total reqs	Users	Avg input tokens	Intra-user reuse	Global reuse
LMSYS-Chat-1M	1,000,000	—	467	—	9.0%
WildChat-4.8M	3,199,860	1,833,730	2,925	5.3%	34.4%
GLM-5.1 production	411,169	4,984	21,043	53.7%	55.3%

- Empirical results (§6) on two consecutive 30-minute production-trace windows. GORGO achieves the lowest TTFT at every percentile (p50, p95, p99) on both a tuning and a held-out evaluation window, against seven baseline policies.

We do not claim any single GORGO variant dominates everywhere: `gorgo-hillclimb` wins all three percentiles in the tuning window, while in the evaluation window `gorgo-static` (deploying the tuning-window learned weights) wins p50 and p95 and `gorgo-hillclimb` wins p99. The W2 p99 margin is at the noise floor of one 30-minute window (§7).

2 Workload Characterization

Routing policies that perform well on short, low-reuse prompts may perform differently on long, high-reuse prompts because the prefill term in TTFT scales with uncached input tokens. We measure three datasets along the same axes: request length, conversational structure, and token-weighted prefix reuse.

Datasets. **LMSYS-Chat-1M** [Zheng et al., 2024a] is the Chatbot Arena dataset of one million conversations. **WildChat-4.8M** [Zhao et al., 2024] is a corpus of 3.2M ChatGPT-proxy conversations grouped by client IP. **GLM-5.1 production trace** is a 7-day chat-completion log from a long-context production endpoint [GLM Team, 2024]. It contains 411,169 requests across 4,984 distinct users, yielding ~ 82 requests per user on average.

We measure prefix reuse with a token-weighted prefix trie that ingests each request’s tokenized message sequence in arrival order, partitioned by user where the dataset carries user identity. The trie reports *intra-user savings* (fraction of tokens matching a prefix from the same user’s prior requests), *cross-user savings* (additional fraction from pooling across users), and *global savings* (sum). LMSYS does not carry user identity, so we report only the global figure.

Three contrasts stand out (Table 1). *Request length*: GLM-5.1 prompts are $45\times$ longer than LMSYS and $7\times$ longer than WildChat. At typical prefill rates, a 21,000-token uncached prompt takes seconds. *User concentration*: GLM-5.1 has 82 requests per user on average; WildChat has 1.7; LMSYS has effectively one. Intra-user reuse depends on returning users, which the public sets simply do

not have. *Reuse magnitude*: 53.7% of GLM-5.1 tokens are part of a prefix seen earlier from the same user (5.3% for WildChat, essentially zero for LMSYS). The bulk of WildChat’s 34.4% global reuse comes from cross-user shared structure (chat templates, viral prompts), not sustained per-user dialogue.

For routing, LMSYS and WildChat live in a regime where prefix-aware policies have little to win. On GLM-5.1, a strategy that lands a user on a replica that has seen their conversation can save more than half of the prefill.

Experimental windows. We focus on two consecutive 30-minute windows of GLM-5.1 traffic: 00:30–01:00 UTC (W1, tuning) and 01:00–01:30 UTC (W2, held-out evaluation). Traces are replayed at original arrival speed and pre-filtered to 24,000 input tokens. Each window contains approximately 4,800 (W1) and 4,200 (W2) requests. Output is capped at 128 tokens so TTFT, not decode, dominates the latency budget.

3 GORGO

3.1 TTFT Decomposition

Let a request r with token sequence x_r be routed to replica u . Its TTFT decomposes as

$$\text{TTFT}(r, u) = \text{rtt}(u) + w_{\text{queue}}(u, t_r) + t_{\text{prefill}}(u) \cdot |x_r \setminus c_u(t_r)| + \varepsilon, \quad (1)$$

where $\text{rtt}(u)$ is the round-trip from the routing proxy to u ; $w_{\text{queue}}(u, t_r)$ is queueing delay; $t_{\text{prefill}}(u)$ is the per-token prefill rate; $c_u(t_r)$ is the set of token sequences cached on u at arrival time t_r ; and ε collects scheduler overhead and batching variance. Cross-region $\text{rtt}(u)$ ranges from 26 ms to 466 ms in our cluster; the prefill term can be the largest and most routing-sensitive term at typical rates of 0.1–1 ms per uncached token on a 21,000-token prompt.

3.2 Cost Model

A GORGO replica score is a weighted sum of the three controllable terms in Equation 1:

$$\text{score}(u, r) = \text{rtt}_u + t_{\text{prefill}} \cdot (|x_r| - |x_r \cap c_u|) + \alpha \cdot (\text{queued_tokens}_u + \text{used_kv_tokens}_u). \quad (2)$$

The proxy sends r to the replica with the minimum score. Three signals feed the score: (i) rtt_u from the EWMA of a lightweight network probe decoupled from the metrics scrape; (ii) $|x_r \cap c_u|$ from the proxy’s local radix trie of cached prefixes per replica, refreshed on every request completion; and (iii) load from the proxy’s in-flight token counter combined with SGLang’s reported KV-cache occupancy. The two free parameters are the per-token prefill rate t_{prefill} and the load weight α .

3.3 Three Variants of Weight Selection

gorgo-static fixes t_{prefill} and α before the run and never adjusts them. In W1 we use manually chosen starter values ($t_{\text{prefill}} = 0.07$, $\alpha = 0.06$). In W2, **gorgo-static** deploys the per-replica weights that **gorgo-hillclimb** ended W1 at ($t_{\text{prefill}} \approx 0.983$, $\alpha \approx 4.4 \times 10^{-4}$) and never adjusts them again. This isolates the contribution of the cost model independent of online adaptation.

gorgo-autotune re-fits both parameters every 16 new request samples by taking the median of the observed rates $(\text{TTFT} - \text{rtt})/\text{uncached_tokens}$ from the trailing 64-sample window, partitioned by destination replica. It adapts to per-replica hardware drift and RTT shifts, but treats the cost-model weights as identifiable physical rates rather than black-box knobs.

gorgo-hillclimb runs a Gaussian (1+1)-evolution strategy [Rechenberg, 1973] over the same two parameters in log-space, with Rechenberg’s 1/5-success rule controlling step size. The objective is the negative p95 TTFT of the rolling 64-request window. Every 16 new samples the tuner scores the current weights, accepts or rejects the previous candidate against the incumbent, and proposes a new candidate by perturbing each parameter as $\exp(\log(v) + \sigma \mathcal{N}(0, 1))$ with σ adapted according to the recent acceptance rate. This variant treats the cost-model weights as black-box knobs: it finds whatever values produce the best p95 TTFT under the current arrival distribution, with no offline profiling.

4 Baselines

We compare GORGO against five baseline policies.

random samples a replica uniformly per request; it is the lower bound.

least-request routes to the replica with the fewest in-flight requests, scoring by $\max(\text{sgLang_running_reqs}_u, \text{proxy_inflight}_u)$ to bridge the ~ 10 s staleness between metrics scrapes [F5 NGINX, 2024].

least-load routes to the replica with the lowest aggregate token-weighted load: $\text{running+queued+used_kv_tokens+queued_tokens}_u$. It is heavier-signal than least-request but cache-blind.

prefix-cache is the AIBrix-style longest-prefix-match policy [AIBrix Team, 2024]. It picks the replica with the longest cached prefix match and falls back to least-request when the running-request imbalance exceeds a soft threshold.

simple-session-affinity hashes the first 256 token IDs of the prompt modulo the number of replicas. The same prefix always maps to the same replica, maximizing intra-user reuse but providing no load-escape mechanism.

5 Experimental Setup

Hardware and software. Each policy runs on its own dedicated three-replica fleet so routing decisions cannot warm or cool another policy’s caches. Each fleet consists of one L40S:2 SGLang [Zheng et al., 2024b] server (tensor-parallel 2, 96 GB VRAM per replica) in each of Asia (Seoul), Europe (Frankfurt), and

US East (Ashburn), serving Qwen3.5-35B-A3B-FP8 [Qwen Team, 2025]. The model’s native context window is 32,768 tokens; we cap workload input at 24,000 tokens to leave KV headroom for in-flight batches. A CPU proxy in US East hosts the routing policy and replays the trace.

The proxy scrapes Prometheus metrics from each replica every ~ 10 s and runs a separate lightweight HTTP probe to estimate per-replica RTT, smoothed with an EWMA. The probe is decoupled from the metrics scrape to avoid polluting the RTT estimate with SGLang handler contention. Before each run, a homogeneity check issues 16 streaming requests directly to each replica, bypassing the proxy; runs with worst-to-median TTFT ratio exceeding $3\times$ are flagged. No such run occurred in these experiments.

Cross-region RTT distribution. Probed from the US East proxy, EWMA-smoothed RTT to the US East replica is 26–40 ms, to Frankfurt is 90–130 ms, and to Seoul is 260–466 ms. The Seoul upper bound reflects routing-table churn during the trace. The overall range of 26–466 ms represents an $18\times$ spread.

Policies and windows. We evaluate eight policies—`random`, `least-request`, `least-load`, `prefix-cache`, `simple-session-affinity`, `gorgo-static`, `gorgo-autotune`, `gorgo-hillclimb`—in each of two windows. All eight fleets start simultaneously at the same wall-clock time with the same input trace. W1 seeds GORGO variants with manually chosen starting weights ($t_{\text{prefill}} = 0.07$, $\alpha = 0.06$). W2 seeds them with the per-replica weights that `gorgo-hillclimb` learned during W1 ($t_{\text{prefill}} \approx 0.983$, $\alpha \approx 4.4 \times 10^{-4}$). `gorgo-static` in W2 deploys exactly those weights and never adjusts them; the other GORGO variants continue tuning on the held-out window. Concurrency is 32 in-flight requests per proxy.

Metrics. For each policy and window we report TTFT p50, p95, and p99 over all successful streaming responses. Failed requests (HTTP 4xx/5xx, dominated by prompt-too-long responses at the model’s effective context boundary) are excluded from latency percentiles but counted in the per-policy success rate. Success rate is between 84.2% and 85.1% across all policies and windows.

6 Results

6.1 W1: Tuning Window

In W1 (Table 2), `gorgo-hillclimb` delivers the lowest p50 (0.288 s), p95 (1.466 s), and p99 (2.307 s). The next-best p95 is `simple-session-affinity` at 1.482 s, a 1.1% gap. The `gorgo-static` configuration—with manually-set starting weights, never adjusted—already beats every non-GORGO policy on p99 and ranks third on p95, showing that the cost model alone improves on single-signal heuristics. The p95 spread across all policies in W1 is $1.25\times$ (1.466 s vs. 1.828 s).

`prefix-cache` is competitive at p95 (1.654 s) but tail-degraded at p99 (3.017 s) because it concentrates traffic onto warm-cache replicas and pays a queueing

Table 2: GLM-5.1 W1 (tuning window, 00:30–01:00 UTC). TTFT in seconds. Bold is best in column. Completed counts from proxy logs; success rate ≈ 84.2 – 84.6% .

Policy	Completed	p50	p95	p99	max	Succ.%
gorgo-hillclimb	4,065	0.288	1.466	2.307	4.39	84.4
simple-session-affinity	4,054	0.329	1.482	2.804	14.96	84.2
gorgo-static	4,076	0.299	1.605	2.357	4.16	84.6
least-request	4,074	0.293	1.632	2.355	13.71	84.6
prefix-cache	4,075	0.297	1.654	3.017	5.96	84.6
gorgo-autotune	4,076	0.294	1.658	2.428	4.18	84.6
least-load	4,066	0.297	1.728	2.629	4.64	84.4
random	4,076	0.296	1.828	3.171	28.99	84.6

Table 3: GLM-5.1 W2 (held-out evaluation window, 01:00–01:30 UTC). **gorgo-static** deploys the W1-learned weights and does not adjust them. Bold is best in column.

Policy	Completed	p50	p95	p99	max	Succ.%
gorgo-static	3,573	0.294	1.510	2.791	15.26	84.9
gorgo-hillclimb	3,583	0.334	1.535	2.595	12.68	85.1
prefix-cache	3,578	0.316	1.632	2.860	34.34	85.0
least-request	3,579	0.303	1.757	2.663	5.22	85.1
random	3,575	0.336	1.763	2.612	30.93	85.0
least-load	3,580	0.363	1.813	2.659	3.84	85.1
gorgo-autotune	3,578	0.368	1.888	3.001	5.70	85.0
simple-session-affinity	3,570	0.396	2.080	4.917	36.60	84.8

penalty when that replica is momentarily saturated. **simple-session-affinity** achieves the second-best p95 on W1 because the production trace has strong per-user temporal locality, but its p99 (2.804 s) and max (14.96 s) reflect episodes where the hashed replica is transiently overloaded with no escape mechanism.

Starting from $t_{\text{prefill}} = 0.07$, $\alpha = 0.06$, the ES converged to $t_{\text{prefill}} \approx 0.983$ and $\alpha \approx 4.4 \times 10^{-4}$ over the 30-minute window. The high learned t_{prefill} reflects the long uncached prefixes in this trace (21k-token prompts, 55% reuse leaves ~ 9.5 k tokens to prefill on average); the low α reflects that SGLang’s continuous batching already orders within a replica and deep queues are rare at 32-concurrency on this arrival rate.

6.2 W2: Held-Out Evaluation Window

In W2 (Table 3), **gorgo-static**—deploying the W1-learned weights without adjustment—achieves the best p50 (0.294 s) and p95 (1.510 s). **gorgo-hillclimb** achieves the best p99 (2.595 s). The two GORGO variants together sweep all three percentiles; no single GORGO variant does. The p95 spread in W2 is

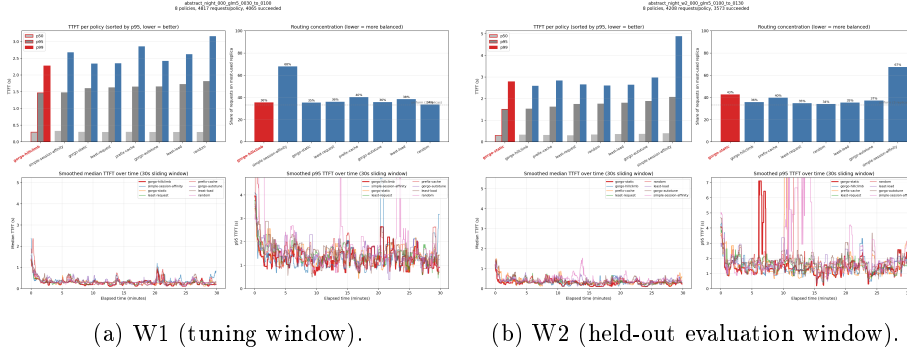


Figure 1: TTFT distributions for all eight policies on the GLM-5.1 trace. **gorgo-hillclimb** sweeps p50/p95/p99 in W1; in W2 **gorgo-static** (W1-learned weights, frozen) leads at p50/p95 and **gorgo-hillclimb** leads at p99.

1.38 $\times$, wider than W1’s 1.25 $\times$.

gorgo-static beats **gorgo-hillclimb** on p50/p95 in W2 because the W1-learned weights are near-optimal for the W2 distribution; continuing to tune on a similar distribution adds variance from each ES proposal. This is a known property of (1+1)-ES under stationary objectives. The practical takeaway is that operators can run **gorgo-hillclimb** during a tuning window and freeze the result into **gorgo-static** for production.

gorgo-autotune underperforms in W2. The median-of-rates fit inverts Equation 1 on trailing samples, but the rate $(\text{TTFT} - \text{rtt})/\text{uncached}$ is unstable when uncached counts are small (high cache hit), inflating the estimate. **gorgo-hillclimb** avoids this because its objective integrates over the mixture rather than trying to disentangle it.

simple-session-affinity’s p95 inflates 40.4% from W1 to W2 and its p99 by 75.4%—a hash-to-replica mapping cannot rebalance when a hash bucket spikes in the second window. By contrast, **gorgo-static**’s W2 p95 (1.510s) improves over its W1 p95 (1.605s), a 6.0% gain; it is the only policy whose p95 improves across windows.

6.3 Summary Figures

7 Limitations

Single-trace generalization. The percentile-sweep claim rests on two consecutive 30-minute windows of one production trace, sharing model, fleet, hour-of-day, and arrival-rate band. W2 demonstrates that tuned weights survive a held-out window of the same workload, not arbitrary transfer across hours, days, or models.

p99 noise floor. Each window has approximately 4,200–4,800 successful responses. p99 over a sample of size n has standard error approximately $\sigma_{\text{TTF}}/\sqrt{n \cdot 0.01 \cdot 0.99} \approx 0.05$ s. The W2 p99 spread between `gorgo-hillclimb` (2.595 s) and `random` (2.612 s) is 0.017 s, inside one standard error. This margin is reported for completeness but should not be interpreted as a reliable advantage. The W1 sweep and W2 p50/p95 wins are several standard errors clear of zero.

Single-tenant traffic. Each policy ran on its own three-replica fleet, so cache state is single-tenant. Multi-tenant sharing changes the queue and cache dynamics that GORGO’s load weight captures; convergence under non-stationary multi-tenant load is not validated here.

Homogeneous hardware and no PD-disaggregation. All replicas use identical GPU configurations. The fleet does not split prefill and decode [Zhong et al., 2024, Patel et al., 2024]; such architectures would require an additional routing term. GORGO uses only metrics exposed by an HTTP-fronted SGLang/vLLM server, not scheduler-internal information [Pan et al., 2025, Yang et al., 2025a] that would tighten the prefill estimate.

ES convergence time. The (1+1)-ES requires several hop cycles to converge from a random starting point. In W1 it converged within the 30-minute window, but very short traffic windows may not provide enough requests.

8 Related Work

Prefix caches and reuse. RadixAttention in SGLang [Zheng et al., 2024b] stores per-request KV state in a radix tree; PagedAttention in vLLM [Kwon et al., 2023] enables prefix sharing through paged memory. KVLink [Yang et al., 2025b], ChunkKV [Liu et al., 2025], KVFlow [Pan et al., 2025], and Learned Prefix Caching [Yang et al., 2025a] improve the single-replica cache but do not decide which replica serves a request.

Routing and scheduling. Mooncake [Qin et al., 2024] is a KV-cache-centric architecture and the source of our trace format. AIBrix [AIBrix Team, 2024] provides prefix-aware load balancing as a baseline; k-LPM [Dexter et al., 2025] formulates LLM scheduling under TTF constraints as NP-hard. DLPM [Cao et al., 2025] targets fairness with locality. Llumnix [Sun et al., 2024] migrates requests across replicas. Multi-LLM routers [Wu and Silwal, 2025] address model selection. CacheBlend [Yao et al., 2024] routes by trie-measured prefix length but does not measure network RTT. DistServe [Zhong et al., 2024] and Splitwise [Patel et al., 2024] disaggregate prefill from decode. GORGO commits to a single scalar score per replica with online weight tuning and requires no engine modifications.

Online hyperparameter adaptation. Bayesian and bandit-based methods have been applied to serving configuration [Wu and Silwal, 2025]. For a 2-dimensional parameter space the (1+1)-ES [Rechenberg, 1973] provides fast, overhead-free convergence with no surrogate model.

9 Conclusion

We characterize a workload regime—long-context, multi-turn, high prefix-reuse production traffic—where standard LLM-serving routing heuristics leave performance on the table because they pick one signal out of three (cache locality, replica load, network cost). On a 411,000-request production trace where 53.7% of input tokens are part of a previously-seen prefix from the same user, three variants of a joint cost model collectively achieve the lowest TTFT at every percentile (p50, p95, p99) on both a tuning and a held-out evaluation window, against seven baseline policies, on a multi-region L40S:2 fleet with cross-region RTTs from 26 to 466 ms. Per-percentile wins are not held by any single fixed weight setting; the best evaluation-window configuration is the one that froze the weights learned in the tuning window. The cost model and tuner fit on a stock HTTP gateway and require no offline profiling.

References

- AIBrix Team. AIBrix: An open-source infrastructure for LLM inference at scale. <https://github.com/vllm-project/aibrix>, 2024.
- Shiyi Cao, Yichuan Wang, Ziming Mao, Pin-Lun Hsu, Liangsheng Yin, Tian Xia, Dacheng Li, Shu Liu, Yineng Zhang, Yang Zhou, Ying Sheng, Joseph Gonzalez, and Ion Stoica. Locality-aware fair scheduling in LLM serving. *arXiv preprint arXiv:2501.14312*, 2025.
- Gregory Dexter, Shao Tang, Ata Fatahi, Qingquan Song, Tejas Dharamsi, and Aman Gupta. LLM query scheduling with prefix reuse and latency constraints. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.
- F5 NGINX. NGINX documentation: Load balancing methods. <https://docs.nginx.com/nginx/admin-guide/load-balancer/http-load-balancer/>, 2024.
- GLM Team. GLM-4: A family of open multilingual multimodal chat LLMs, 2024.
- Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, and Ion Stoica. Efficient memory management for large language model serving with PagedAttention. In *ACM Symposium on Operating Systems Principles (SOSP)*, 2023.

- Xiang Liu, Zhenheng Tang, Peijie Dong, Zeyu Li, Yue Liu, Bo Li, Xuming Hu, and Xiaowen Chu. ChunkKV: Semantic-preserving KV cache compression for efficient long-context LLM inference. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.
- Zaifeng Pan, Ajikumar Patel, Zhengding Hu, Yipeng Shen, Yue Guan, Wan-Lu Li, Lianhui Qin, Yida Wang, and Yufei Ding. KVFlow: Efficient prefix caching for accelerating LLM-based multi-agent workflows. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. arXiv:2507.07400.
- Pratyush Patel, Esha Choukse, Chaojie Zhang, Aashaka Shah, Íñigo Goiri, Saeed Maleki, and Ricardo Bianchini. Splitwise: Efficient generative LLM inference using phase splitting. In *ACM/IEEE International Symposium on Computer Architecture (ISCA)*, 2024.
- Ruoyu Qin, Zheming Li, Weiran He, Mingxing Zhang, Yanzhe Wu, Weimin Zheng, Xinran Chen, Rui Xu, Xingda Yao, Rongxin Wei, Zhiyuan Zhao, Jing Huang, Mingjie Yang, and Jidong Wu. Mooncake: A KVCache-centric disaggregated architecture for LLM serving, 2024.
- Qwen Team. Qwen3 technical report. 2025. Alibaba Cloud.
- Ingo Rechenberg. Evolutionsstrategie: Optimierung technischer systeme nach prinzipien der biologischen evolution. 1973.
- Biao Sun, Ziming Huang, Hanyu Tang, Chengquan Wang, Cheng Zhang, Yiming Li, Liu Yang, Wencong Zhang, Lin-Wei Wang, Tianhe Wu, Ang Cao, Yongqiang Lin, et al. Llumnix: Dynamic scheduling for large language model serving. In *USENIX Symposium on Operating Systems Design and Implementation (OSDI)*, 2024.
- Fangzhou Wu and Sandeep Silwal. Efficient training-free online routing for high-volume multi-LLM serving. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.
- Dongsheng Yang, Austin Li, Kai Li, and Wyatt Lloyd. Learned prefix caching for efficient LLM inference. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025a.
- Jingbo Yang, Bairu Hou, Wei Wei, Yujia Bao, and Shiyu Chang. KVLink: Accelerating large language models via efficient KV cache reuse. *arXiv preprint arXiv:2502.16002*, 2025b.
- Jiayi Yao, Hanchen Li, Yuhao Liu, Siddhant Ray, Yihua Peng, Hao Zhang, Ion Stoica, Kurt Keutzer, and Michael W. Mahoney. CacheBlend: Fast large language model serving for RAG with cached knowledge fusion, 2024.
- Wenting Zhao, Xiang Ren, Jack Hessel, Claire Cardie, Yejin Choi, and Yuntian Deng. WildChat: 1M ChatGPT interaction logs in the wild. In *International Conference on Learning Representations (ICLR)*, 2024.

- Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Tianle Li, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zhuohan Li, Zi Lin, Eric P. Xing, Joseph E. Gonzalez, Ion Stoica, and Hao Zhang. LMSYS-Chat-1M: A large-scale real-world LLM conversation dataset. In *International Conference on Learning Representations (ICLR)*, 2024a.
- Lianmin Zheng, Liangsheng Yin, Zhiqiang Xie, Chuyue Livia Sun, Jeff Huang, Cody Hao Yu, Shiyi Cao, Christos Kozyrakis, Ion Stoica, Joseph E. Gonzalez, Clark Barrett, and Ying Sheng. SGLang: Efficient execution of structured language model programs. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024b.
- Yinmin Zhong, Shengyu Liu, Junda Chen, Jianbo Hu, Yibo Zhu, Xuanzhe Liu, Xin Jin, and Hao Zhang. DistServe: Disaggregating prefill and decoding for goodput-optimized large language model serving. In *USENIX Symposium on Operating Systems Design and Implementation (OSDI)*, 2024.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [\[Yes\]](#)

Justification: The abstract claims (i) public datasets underrepresent long-context multi-turn traffic (quantified in §2), (ii) we evaluate routing policies on a production trace (Tables 2 and 3), and (iii) GORGO sweeps every TTFT percentile in both windows. We are explicit in the introduction and §7 that “GORGO” refers to a family and that the W2 p99 margin is at the noise floor.

2. Limitations

Question: Does the paper discuss the limitations of the work?

Answer: [\[Yes\]](#)

Justification: Section 7 lists single-trace generalization, the W2 p99 noise floor, the single-tenant assumption, `gorgo-autotune` instability, hardware homogeneity, the absence of PD-disaggregation, and ES convergence time.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete proof?

Answer: [\[N/A\]](#)

Justification: The paper presents an additive cost model and an online optimizer with no formal theorems.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results?

Answer: [\[Yes\]](#)

Justification: Section 5 specifies model, regions, hardware, fleet topology, concurrency, trace windows, and per-window GORGO seeding. Section 3 gives Equation 2, the (1+1)-ES configuration ($\sigma_0 = 0.5$, 1/5-rule, hop 16, window 64), and the median-of-rates fit. The reproducibility statement enumerates four canonical paths in the released artifact.

5. Open access to data and code

Question: Does the paper provide open access to the data and code?

Answer: [\[No\]](#)

Justification: The GLM-5.1 production trace is not redistributable under its terms of service. The proxy and policy code is not yet open-sourced at submission time. The public-dataset characterization (LMSYS, Wild-Chat) is fully reproducible with no proprietary inputs.

6. Experimental setting/details

Question: Does the paper specify all the training and test details?

Answer: [\[Yes\]](#)

Justification: Section 5 specifies model, hardware, regions, concurrency, trace windows, max input tokens, and per-window seeding for adaptive policies. ES configuration is given in Section 3.

7. Experiment statistical significance

Question: Does the paper report error bars or appropriate statistical significance information?

Answer: **[No]**

Justification: Each data point is a single 30-minute window. Section 7 derives an approximate p99 standard error and shows that the W2 p99 spread between `gorgo-hillclimb` and `random` is inside one standard error. The W1 sweep and W2 p50/p95 wins are several standard errors clear of zero.

8. Experiments compute resources

Question: Does the paper provide sufficient information on compute resources?

Answer: **[Yes]**

Justification: Each policy runs on a dedicated 3-replica fleet of L40S:2 SGLang servers (tensor-parallel 2, 96 GB VRAM per replica). Eight policies $\times$ two windows = 16 fleet-runs, each $\sim$ 30 min wall clock, on three regions. Total GPU-hours: $\approx$ 24.

9. Code of ethics

Question: Does the research conform with the NeurIPS Code of Ethics?

Answer: **[Yes]**

Justification: The work concerns routing infrastructure. The production trace is used under terms that permit research replay; we release only aggregated prefix-trie statistics, not individual requests. No individuals are identifiable from the released aggregates.

10. Broader impacts

Question: Does the paper discuss potential societal impacts?

Answer: **[Yes]**

Justification: Routing improvements reduce the energy and dollar cost of serving LLMs at fixed quality. Lower-cost serving may accelerate deployment in settings where that is socially harmful; the contributions are infrastructure-level and do not affect model capability.

11. Safeguards

Question: Does the paper describe safeguards for responsible release?

Answer: **[N/A]**

Justification: We do not release datasets or models. The released code is a routing proxy and policy library with no inherent dual-use beyond LLM-serving infrastructure.

12. Licenses

Question: Are creators or owners of assets properly credited?

Answer: [\[Yes\]](#)

Justification: SGLang, vLLM, AIBrix, LMSYS-Chat-1M, WildChat-4.8M, and Qwen3 are cited and used under their published licenses.

13. **New assets**

Question: Are new assets introduced in the paper well documented?

Answer: [\[Yes\]](#)

Justification: The proxy, policy library, experiment controller, and trace builder are documented in the repository’s top-level documentation. Spec and manifest files are stored alongside the controller.

14. **Crowdsourcing and research with human subjects**

Question: Does the paper involve crowdsourcing or human subjects?

Answer: [\[N/A\]](#)

Justification: No human subjects.

15. **IRB approvals**

Answer: [\[N/A\]](#)

Justification: No human subjects.

16. **Declaration of LLM usage**

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the research methodology?

Answer: [\[N/A\]](#)

Justification: LLM serving infrastructure is the object of study. Authors used LLMs for routine writing assistance only.

Reproducibility Statement

The experiments are driven by a single experiment controller that launches fleets, configures policies, and replays a trace per policy. The four canonical paths in the released artifact are:

- `specs/policy_matrix_abstract_night.json` — W1 spec (eight policies, manual GORGO starter weights);
- `specs/policy_matrix_abstract_night_w2.json` — W2 spec (eight policies, GORGO seeded with W1-learned weights);
- `experiment_runner/policy_matrix_app.py` — experiment controller;
- `data_processing/build_mooncake_trace.py` — trace builder.

W1 and W2 manifests, result CSVs, and summary plots are produced by running the controller against these specs and post-processing per-policy stats with the released analysis scripts. The GLM-5.1 production trace is not redistributable; the public-dataset characterization (LMSYS, WildChat) reproduces with no proprietary inputs.